

Large-Neighborhood Search as Audited Plan Improvement

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Abstract

This paper documents the BISECT local-search and large-neighborhood-search improvement contract. It treats `bisect improve` as a lineage-preserving heuristic implementation: improved plans, rejected moves, no-op outcomes, and staged LNS/tabu paths must all produce reproducible summaries. Accepted improvements must produce RPLAN/RCTX descendants that name the parent plan, child artifact, method, status, objective delta, and validation profile.

1 Introduction

U.18 documents `bisect-local-search` and `bisect-cli::improve_cmd`, exposed as `bisect improve` with staged `--search lns` and `--search tabu` surfaces. The paper is a heuristic improvement implementation paper, not a global optimality claim.

The method is a descendant-plan workflow. It starts from an audited plan and context, evaluates local moves against the same profile constraints, and writes a child package only when the accepted assignment still verifies. This places U.18 between the construction papers and U.20: construction creates a starting plan, local search explains an improvement transition, and the audit certificate fixes the child plan as a reusable artifact.

2 Implementation Contract

The current implementation centers on a one-move boundary improvement baseline plus structured statuses for no-improvement, rejected-move, and staged-method paths. The summary must record parent plan id, attempted method, accepted move or rejection/no-op reason, objective delta, validity checks, and output artifact id when a child plan is emitted.

The atlas framing treats every boundary move as a candidate that must pass separate gates. A move that improves the edge-cut objective can still be rejected for population or connectedness, and a no-op run is meaningful only if the evaluated neighborhood and rejection reasons are recorded. The descendant package is therefore evidence of a local transition from a parent plan, not a new construction from scratch.

The L0 improvement fixture makes the transition small enough to inspect. The starting assignment is $[0, 0, 0, 1, 1, 1]$ on a six-unit graph with equal populations and two districts. Unit 2 touches the opposite district through units 3, 4, and 5. Moving unit 2 from district 0 to district 1 preserves population tolerance at 34 percent, keeps both districts connected, and reduces edge cut from 3 to 1. The emitted child assignment is $[0, 0, 1, 1, 1, 1]$. A competing move that improves edge cut but breaks population or contiguity would be recorded as evaluated and rejected, not silently treated as an improvement.

Outcome	Required summary	Claim supported
Improved	parent, child, move, delta, validity checks	local improvement
Rejected move	parent, candidate move, rejection reason	validity-preserving search
No-op	parent, search status, reason	reproducible non-improvement
Staged method	requested method and unsupported scope	roadmap boundary

Table 1: Improvement-family statuses.

Summary field	Current value family	Interpretation
Summary schema	local-search summary v1	Stable JSON envelope for transition evidence
Method	one-move, staged tabu, staged LNS	Search surface requested
Status	improved or no-improvement	Whether a child assignment changed
Move counts	evaluated and accepted counts	Why no-op runs are inspectable
Edge cut	initial and final values	Objective delta evidence
Population deviation	initial and final values	Validity gate evidence
Parameter hash	SHA-256 digest	Tamper check for summary-defining inputs

Table 2: Concrete local-search summary fields emitted by `bisect-local-search`.

The paper therefore does not treat every command invocation as a successful improvement. A no-op transcript is evidence that the search was run under a declared method and profile. A rejected-move transcript is evidence that the implementation preserved validity by refusing a candidate. Only an *Improved* status supports the claim that a descendant plan was emitted with a measured local objective delta.

3 Audit Artifacts

Improved plans are descendants of an existing RPLAN input. The RCTX and audit certificate should preserve parent/child lineage, declared profile checks, and the improvement summary so a verifier can inspect both validity and provenance. The audit package must identify the parent RPLAN, emitted child RPLAN when one exists, method, status, objective before and after, objective delta, validation profile, and any random seed used by the method.

Validity and improvement are separate claims. A valid child plan is not automatically a meaningful improvement, and an apparent objective improvement is not accepted unless the declared profile checks pass. No-op and staged-method records still belong in the audit trail because they explain why no descendant artifact was produced.

The public golden package can be checked either through the RPLAN certificate verifier or through the BISECT manifest verifier.

The parent-child chain is the audit-critical object. The CLI computes the parent plan hash, writes `local-search-summary.json`, stores its digest in algorithm lineage, and emits `improved.rplan`, `improved.rctx`, `audit-certificate.json`, and `manifest.json`. The certificate verifies the child plan; the summary explains the transition from the parent. That separation prevents an edge-cut improvement from being confused with a legal-validity claim or a global-search claim.

4 Evaluation Plan

Current evidence is L0 behavior for improvements, no-ops, rejected moves, and staged methods; L1 CLI and RPLAN sidecar tests; and a public U.18 parent/child lineage package in the package corpus. A benchmark-tier 100-unit synthetic grid package now records parent-plan lineage and no-improvement behavior over a larger audited input. Empirical distributions of accepted moves, objective deltas, and tabu/LNS comparisons are future evidence.

Claim	Current evidence	Publication gap
One-move search can emit valid descendants	L0 improvement fixtures, L1 RPLAN sidecar tests, public lineage package, and 100-unit benchmark package	larger real-data before/after package
No-op and rejected moves are reproducible	L0 status fixtures	CLI transcripts for each status
Staged LNS/tabu surfaces are honest roadmap boundaries	Staged-method status tests	Full method implementation
Large-instance heuristic quality improves materially	Future evidence only	Benchmark protocol and comparison baselines

Table 3: Claim and evidence ladder for U.18.

The representative CLI shape is:

Command or package	Evidence produced
<code>cargo test -p bisect-local-search</code>	Checks deterministic one-move improvement, no-improvement, staged-method errors, summary hashes, and lineage.
<code>bisect improve</code> fixture run	Writes the local-search summary, child RPLAN/RCTX, audit certificate, manifest, and parent-plan lineage.
Public U.18 packages	Preserve improvement, generated-descendant, and grid10 benchmark/no-improvement examples in the package corpus.

Table 4: Current U.18 evidence path.

```
bisect improve --method one-move \
  --plan input.rplan --context input.rctx \
  --out-dir out --tolerance 34
```

The grid10 benchmark package records the equally important no-improvement case: the search ran, preserved the parent assignment, and still emitted auditable summary and lineage artifacts.

5 Limitations

One-move local search can get stuck in local optima and should not be reported as global search. Advanced LNS and tabu methods are scaffolded until their performance protocols and validity-preserving repair hooks are documented. The current paper supports an audited improvement contract, not a claim that the implementation finds globally optimal or even consistently high-quality plans across large instances.

The future empirical paper must include method parameters, random seeds where relevant, acceptance rules, distributions of accepted moves and objective deltas, and comparisons against no-op, one-move, tabu, and LNS baselines before making broader performance claims. The current benchmark package is synthetic and mainly verifies packaging scale and no-improvement reporting. It should not be read as evidence that one-move search consistently improves real plans or that staged tabu/LNS options are ready for quality comparison.

6 Conclusion

U.18 makes plan improvement auditable: the output is not just a better score, but a child plan with a recorded parent, method, status, objective delta, and verifier path. The paper now treats no-op, rejected, and staged paths as evidence-bearing outputs while reserving LNS/tabu performance claims for a later benchmarked implementation.

References

BISECT project. U.18 large-neighborhood search paper plan, 2026. Local planning document and implementation roadmap.